

DATABASE AND INTERFACES

CW (Design, Implement Database with its interface)

Introduction

The aim of this course is to learn about databases and interfaces. The lectures and the labs are designed to support your technical skills to achieve this goal. The grades are distributed as following:

1. 40% In class quiz
2. 60% Coursework

Submission: **28th April 2025 23:59:00**

Section A: System requirements

ABB Robotics is one of the world's leading providers of robotics and automation solutions. Your task is to develop an online **Customer Relationship Management (CRM)** system for ABB Robotics. The CRM will allow users to create accounts, log in, and access different features based on their roles as Admin or Sales Representative. Admin users will have full control over managing customer records, leads, and user accounts, while Sales Representatives will be able to access and manage their assigned customers and leads. The goal is to build a user-friendly application that allows the company to efficiently manage customer information, track interactions, and organize leads.

The system must have the following functionalities:

a. Customer Management

Each customer will have a profile with important details like name, company, email, phone number, address, and interaction history. This helps ABB Robotics track customer information and manage relationships effectively. Users can add new customer profiles, update details, and quickly search customer information when needed. This ensures accurate and up-to-date records for better service.

b. Lead Management

The system will allow users to manage potential customers or leads effectively. Users can add new leads, update their details, and monitor their progress effectively. Each lead will have a status (e.g., New, Contacted, In Progress, Closed) to indicate its current stage, allowing Sales Representatives to manage their activities efficiently. Users can assign follow-up dates and update lead notes to ensure timely follow-up actions.

c. Interaction History

The system will allow Sales Representatives to record interactions with customers, such as calls, meetings, or emails. This helps keep track of past communications and ensures smooth follow-ups. Users can enter the date, type of interaction, and a short description. The system will also allow searching of interactions, making it easy to find past records when needed.

Advanced Features (Optional)

d. Follow-up Reminders & Notifications

To ensure timely follow-ups with customers and leads, the system can include a reminder and notification feature. Sales Representatives can set reminders for follow-ups, ensuring they never miss important interactions. The system will send automated email notifications for scheduled follow-ups. This will help users stay organized, improve customer engagement, and increase the chances of closing deals.

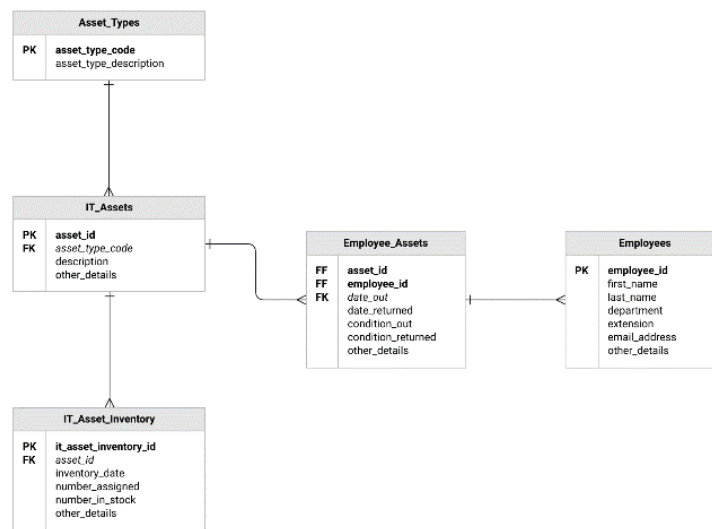
To build this system, you will need to design and implement MySQL database tables to store information such as customer profiles, leads, user accounts, and interaction history. Carefully consider the relationships between the data entities and define primary keys, foreign keys and other necessary constraints. The front-end user interface should be developed using HTML, CSS, and JavaScript to ensure a smooth and user-friendly experience. Server-side scripting with PHP is required to handle authentication, data retrieval, and dynamically managing customer records, leads, and interactions. Form validation techniques should be implemented to prevent users from entering invalid values. Proper documentation and coding best practices should be followed to ensure maintainability and scalability of the system.

Section B: Deliverables

This section outlines the required files to submit for the assignment and how they should be organised.

ER diagram

Each group must submit an ER diagram that visually represents the logical structure of the database. The diagram should use **crow's foot notation**, as shown in the example below to illustrate entities, attributes, relationships, cardinalities, and constraints. Save this as a **PDF file** named "**COMP1044_ERD.pdf**".



Database

To implement the ER diagram, each group must create tables based on the design provided in the ER diagram. After creating the tables, the group should insert sample data into each table to populate the database. Once the tables are created and populated, the entire database must be exported as an SQL file named "**COMP1044_Database.sql**" for submission.

Program Code

Submit the full source code of the system in a **compressed ZIP folder** named "**COMP1044_SRC.zip**". Ensure this folder includes all necessary files such as scripts, images, and dependencies required to run the system. Maintain a clear and structured file organization to facilitate easy testing.

Video demonstration

Each group must record a **video demonstration**, with a maximum duration of **5 minutes**, showcasing the system's functionality and its interaction with the database. The video should clearly present all system functions, including inserting, updating, and querying data, to verify that the system operates correctly. Save the video in **MP4 format** and name it "**video.mp4**".

Work Breakdown Structure

Submit a document detailing the work breakdown structure of the project. This document should outline the individual effort in development tasks and describe each member's contributions. Save this as a PDF file named "**COMP1044_WBS.pdf**".

Folder and File Organisation

Organize all five files (**COMP1044_ERD.pdf**, **COMP1044_database.sql**, **COMP1044_SRC.zip**, **COMP1044_WBS.pdf**, **video.mp4**) into a folder named "**COMP1044_CW_Gx**" (where "x" is the group number). Compress this folder into a **ZIP file** named "**COMP1044_CW_Gx.zip**" and ensure the file size is under **250MB**.

Upload the final **ZIP file** to **Moodle** to complete your coursework submission.

Note: We would like to kindly remind you that our institution employs plagiarism detection software to review your coursework submissions. Plagiarism is strictly prohibited, and any academic misconduct will be addressed following university policies and the guidelines outlined in the student handbook. You are encouraged to engage in informal discussions with fellow students regarding the coursework. However, it is important that your submission reflects your own original work. Copying and pasting text from external sources without proper referencing is not permitted. If you have any uncertainties or questions about the referencing process, please don't hesitate to reach out to the module convenors during our lab sessions or at the conclusion of a teaching session.

MARKING CRITERIA/RUBRIC

Assessment Criteria and Marking Overview Tasks	Marks given	Marks awarded
1. ER diagram • Proper use of entities, attributes, relationships, and cardinalities • Logical structuring of database components • Correct use of crow's foot notation • Clarity and proper labeling	15	
2. Database implementation • Efficient table structures • Use of primary keys, foreign keys, and constraints • Referential integrity enforced • Logical and meaningful sample data	10	
3. Demonstration video • Covers all required features • Explains the key technical elements • Timing (5 minutes)	10	
4. Front-End Development • Clean, responsive, and user-friendly UI (HTML, CSS) • Consistency across pages • Input validation and error handling implemented	15	
5. Backend and Integration • Proper use of PHP for database communication • Smooth interaction between front-end and database for fetching/updating data	15	
6. Functionality • All required features are implemented • The system runs smoothly without errors, crashes, or unexpected behavior • Implementation of extra features beyond the requirement • Unique solution	20	
7. Individual contribution • Clear evidence of individual effort in development tasks	15	
Total marks	100	

Work Breakdown Structure

Students name Task			
Front-End Development			
Customer Management			
Lead Management			
Interaction History			
Back-End Development			
Customer Management			
Lead Management			
Interaction History			
Signature:			